

Aarya Shah

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TECHNICAL SKILLS

Languages: Python, SQL, R, Java, JavaScript, Typescript, C++, C#/.NET, Golang
Tools: Azure, GCP, AWS, Snowflake, Terraform, Redis, Git, Docker, Kubernetes, PostgreSQL
Frameworks: PyTorch, TensorFlow, Sklearn, Langchain, Kafka, Hadoop, Spark, DBT, Airflow

EXPERIENCE

Cognixion September 2025 – December 2025
Machine Learning Software Engineer Santa Barbara, CA

- Launched a memory system in AWS (API Gateway, Lambda, Titan) improving contextual relevance by 30%
- Built multi-agent systems using MCPs & OpenAI Agents SDK, increasing from 500 to 2,000+ concurrent-users
- Conducted A/B testing for event-driven agentic-AI app on S3, webhooks & LangGraph; boosted autonomy by 60%

University of Waterloo August 2025 – December 2025
AI/ML Research Assistant Waterloo, ON

- Co-authored research on efficient Transformer/BERT scaling, reducing FLOPs by 10% via quantization
- Fine-tuned LLaMA-3 8B using QLoRA on status-quo datasets improving F1 by 40% over zero-shot baseline
- Constructed climate-risk classifier in Python (scipy, tensorflow, SVM/GBM, CNNs) for sub-200ms recognition

Globys May 2025 - August 2025
Machine Learning Software Engineer Seattle, WA

- Utilized indexing for NoSQL (Azure) in C# REST APIs to optimize backend & reduce query latency by 70%
- Deployed AI-powered SaaS products via MLOps pipelines for enterprise clients serving 50k+ monthly API calls
- Managed ML lifecycle with MLflow, deploying a 0.88-AUC XGBoost churn-forecast model with drift monitoring

Health Canada September 2024 – December 2024
Data Scientist Toronto, ON

- Implemented NLP-based query complexity screener, reducing compute costs by \$15k+ annually on Databricks
- Trained LSTM & regression forecasting models, improving prediction accuracy by 8% for analytics workflows
- Enforced a feature store in GCP that improved machine learning model scalability & performance by over 25%

IBM July 2023 – August 2023
Machine Learning Engineer Toronto, ON

- Engineered a caching layer with PostgreSQL optimizing load times reducing latency for 4,000,000+ users
- Orchestrated RAG pipeline using Langchain with Pinecone/FAISS vector stores cutting response times by 20%
- Developed file processors (OCR) in C++ & Docker which improved RAG model performance for 10+ file types

RBC May 2022 – August 2023
Data Engineer Mississauga, ON

- Migrated data platform from DB2 to Snowflake, improving scalable analytics & query performance by 20%
- Accelerated data pipelines by validating 25+ data sources in Data Lake, Kafka, Hadoop (Hive) & SQL
- Deployed 20+ data pipelines (ETL) & 10+ CI/CD workflows to Azure using Python, Spark, Airflow & Git

PROJECTS

Quant Platform 🐙 | Python, C++, Golang, Node.js, React, TensorFlow, HTML/CSS

- Deployed full-stack quant trading platform with 45 REST endpoints, quant models, backtests and real-time data
- Engineered fault-tolerant 84 feature ML pipeline with auto fallbacks, TTL caching, and Docker deployment
- Built sub-10µs C++ execution engine (pybind11) & parallel Go risk service handling 50k+ msgs/sec over gRPC

Multimodal Inference Optimization 🐙 | HuggingFace, VLMs, TensorRT-LLM, CUDA, vLLMs

- Improved GPU usage by 15% reducing inference overhead via optimized memory allocation & parallel execution
- Built OpenAI-compatible serving API with adaptive SM control; cuts p99 inference latency from 850ms to 190ms

Policy Q/A chatbot 🐙 (10000+ users) | Python, Typescript, Langchain, HuggingFace, LLMs

- Architected RAG-based insurance Q/A system enabling real-time inference with 750ms median response latency
- Increased RAG faithfulness, relevancy & recall on 500+ queries by moving from naive chunking to BM25

EDUCATION

University of Waterloo September 2021 – April 2026
Bachelor of Mathematics in Statistics & Computer Science Waterloo, ON

- **Relevant Coursework:** Neural Networks, Data Types & Structures, Linear Models, Stochastic Processes